

## SMARTBRIDGE - Project Executable Files

|               |                                             |
|---------------|---------------------------------------------|
| Date          | 25 June 2026                                |
| Team ID       | XXXXXX                                      |
| Project       | Emotion Detection & Learning Support Engine |
| Maximum Marks | 3 Marks                                     |

This document contains all executable and deployable files required to run the Emotion Detection & Learning Support Engine project.

### Step 1 - Submission Checklist

| S.No | Item                   | Submitted              |
|------|------------------------|------------------------|
| 1    | Complete source code   | Yes                    |
| 2    | README / Setup Guide   | Yes                    |
| 3    | requirements.txt       | Yes                    |
| 4    | Database schema / seed | NA (CSV-based storage) |
| 5    | .env.example           | Yes                    |
| 6    | Deployed URL           | Yes                    |
| 7    | APK / Executable       | NA                     |
| 8    | Dockerfile             | NA                     |
| 9    | Test files/results     | Yes                    |
| 10   | Demo Video             | Yes                    |

## Step 2 - File / Folder Structure

```
Emotion-Detection-Learning-Support-Engine/  
■■■■ app.py  
■■■■ preprocessing.py  
■■■■ emotion_detector.py  
■■■■ groq_integration.py  
■■■■ emotion_models.py  
■■■■ history.csv  
■■■■ requirements.txt  
■■■■ README.md  
■■■■ setup_models.py  
■■■■ render.yaml  
■■■■ test_hang.py  
■■■■ .env.example  
■■■■ models/  
■■■■ pages_previews/  
■■■■ reference_img/  
■■■■ reference_text/  
■■■■ assets/
```

## Step 3 - Deployment / Access Details

| Field      | Details                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------|
| Hosted URL | <a href="https://emotion-learning-assistant.streamlit.app">https://emotion-learning-assistant.streamlit.app</a> |
| Login      | No login required – Open Access                                                                                 |
| Platform   | Streamlit Community Cloud                                                                                       |
| Repository | <a href="#">GitHub Repository Link</a>                                                                          |
| Demo Video | <a href="#">Demo Video Link</a>                                                                                 |

#### Step 4 - Run Instructions

1. Install Python 3.10 or above
2. Clone the GitHub repository
3. Navigate to project folder
4. `pip install -r requirements.txt`
5. Create `.env`:  
`GROQ_API_KEY=YOUR_API_KEY`
6. `streamlit run app.py`
7. Open localhost URL
8. Select study field
9. Enter student query
10. Click Analyze & Get AI Help
11. View detected emotion, AI response and analytics.

#### Step 5 - Known Issues / Limitations

| S.No | Issue                                            | Workaround                 |
|------|--------------------------------------------------|----------------------------|
| 1    | Single-user session support only                 | Multi-user support planned |
| 2    | Emotion detection limited to predefined emotions | Future deep learning model |
| 3    | Internet required for Gemini API                 | Offline fallback available |
| 4    | CSV storage                                      | PostgreSQL planned         |
| 5    | No voice emotion detection                       | Planned                    |